Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 27)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 1 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

0 1 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

0 1 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

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0 1 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

0 1 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

0 2 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

0 2 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 2 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale .....

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Large blue butterfly .....

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10

0 3 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 3 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

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Answer

0 3 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

0 3 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

0 3 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

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0 3 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

15

0 4 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 4 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

0 4 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 4 . 3 The test gave a probability p = 0.30. State what this means and what the student should conclude.

[2 marks]

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0 4 . 4 Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

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0 4 . 5 Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 5 . 1 Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 5 . 2 Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

0 5 . 3 Explain **three** ways in which coral reefs are important to humans.

[3 marks]

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0 6 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 6 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 6 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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5

0 7 Table 1 shows some designated protected areas in the UK.

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0 7 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature
Site of Special Scientific Interest (SSSI)		
Special Protection Area (SPA)		
Ramsar site		
Marine Conservation Zone (MCZ)		

0 7 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

0 7 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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0 8 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 8 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 8 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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5

0 9 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 9 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 9 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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1 0 . 2 State what is meant by a **taxon**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS